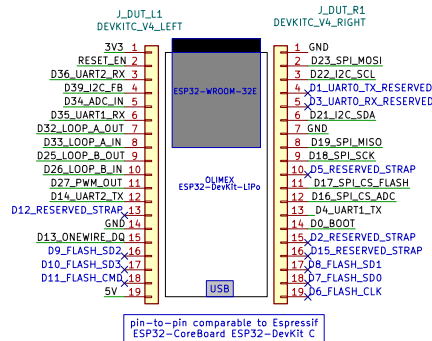
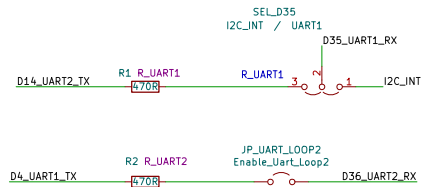


## ESP32-DEVKITC V4 DUT SOCKET / HEADER MAP

Two 1x19 sockets represent the DevKitC V4 headers, viewed from the component side with USB at the bottom.  
Header labels are the electrical definition:



## UART – Full LOOPBACK



## RESET / BOOT AUTOMATION PROVISION



### FIXED GPIO ALLOCATION

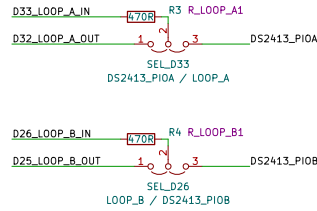
GPIO LOOP A: D32 -> SEL\_D33 -> 470R -> D33  
GPIO LOOP B: D25 -> SEL\_D26 -> 470R -> D26  
PWM/ADC: D27 -> 10K/100nF -> D34 and MCP3008 CH0  
I2C: SDA D21, SCL D22, INT via SEL\_D35 to D35, FB to D39  
SPI: SCK D18, MISO D19, MOSI D23, ADC CS D16, FLASH CS D17  
ONWIRE: D13, DS18B20 bus and removable DS2413  
UART0: D1/D3 reserved for REPL and flashing  
UART TEST: D4 -> JP\_UART\_LOOP2 -> D36  
D14 -> SEL\_D35 -> D35

### DEFAULT SELECTOR / SHUNT STATE

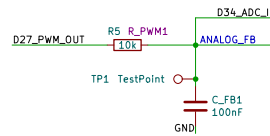
SEL\_D33: pins 1-2 fitted - D32 to D33 GPIO loop  
SEL\_D26: pins 1-2 fitted - D25 to D26 GPIO loop  
SEL\_D35: pins 1-2 fitted - I2C\_INT to D35  
JP\_UART\_LOOP2: OPEN  
JP\_External\_5V1: OPEN  
For UART crosslink testing:  
SEL\_D35 pins 2-3 fitted and JP\_UART\_LOOP2 CLOSED.

D0 reserved for BOOT.  
No fixed loads on strapping pins D2/D5/D12/D15.  
D6-D11 are module flash signals and must not be used.  
D34-D39 are input-only and have no internal pulls.

## LoopBack Test Selectors – GPIO LOOPBACKS / DS2413 Breakout



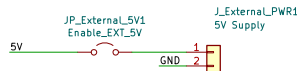
## PWM / ADC FEEDBACK



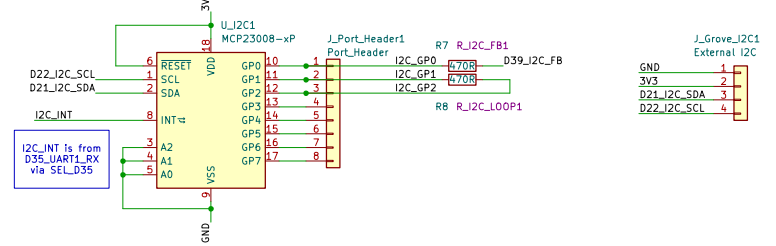
## UART2 – External Peer



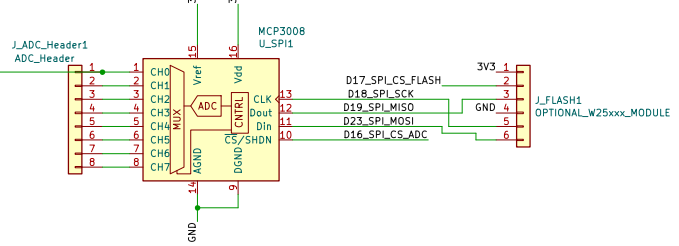
## External 5V Supply



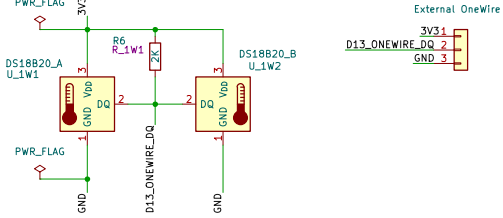
## I2C MCP23008 + GROVE I2C Extension



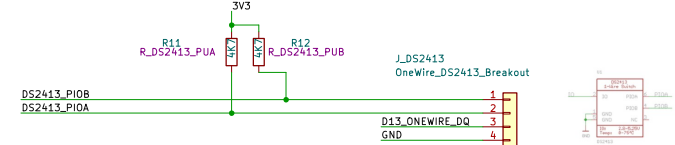
## SPI MCP3008 + OPTIONAL FLASH



## ONEWIRE – TWO DS18B20 + External Option



## ONEWIRE – DS2413 Breakout



UART0 D1/D3 and boot straps remain reserved  
Tests use Espruino Dxx names matching GPIO numbers  
ESP-IDF 5 port test harness – fixed GPIO allocation  
Simon Andrews

Sheet: /  
File: ESP32\_V1.kicad\_sch

Title: Espruino ESP32 DevKitC V4 Test Harness

Size: A3 Date: 2026-06-21  
KiCad E.D.A. 9.0.7

Rev: 1.0  
Id: 1/1